

Volume 3: Web3 Wallet, Solana USDT Ledger & Transactions

Technical specification for Solana blockchain deposit monitoring, append-only double-entry financial ledger, escrow locking, tournament prize payouts, and non-custodial reconciliation.

Target Stack: Solana Web3.js · SPL Token USDT · Prisma Ledger · Redis Balance Gateway · **Lead:** Muhammad Sameer Ali

VOLUME OBJECTIVE:

Specifies the immutable financial engine powering Countdown 31. All real-world currency movement uses Solana SPL Token USDT (6 decimals, 1 USDT = 1,000,000 base units). Details deposit binding, RPC listeners, double-entry ledger bookkeeping, prize pool escrow, and withdrawal safety thresholds.

1. Solana Wallet Integration & Deposit Binding Engine

- **Deposit Address Binding:** In DepositBinding, an external Solana source address is bound uniquely to one user (address @unique). This ensures a deposit transaction on-chain is credited strictly to the verified owner.
- **RPC Transaction Listener:** Background worker subscribes to platform treasury wallet address via Solana WebSocket RPC (onLogs / onAccountChange). When an SPL USDT transfer is finalized (commitment: 'confirmed'), worker parses sender address, matches against DepositBinding, and executes an atomic ledger credit.
- **Base Units Math:** Financial values are stored as 64-bit BigInts (6 decimal places). $\$100.00 \text{ USDT} = 100\,000\,000 \text{ base units}$.

2. Append-Only Double-Entry Ledger System

Transaction Type	Ledger Debit / Credit Flow	Ref Type & Reversible Checks
DEPOSIT	Treasury Wallet (+) / User Balance (+)	SOLANA_SPL_TX (Verified on-chain signature)
WITHDRAWAL	User Balance (-) / Treasury Wallet (-)	MANUAL_REVIEW or Automated SPL Transfer
ENTRY_FEE	User Balance (-) / Tournament Escrow (+)	TOURNAMENT_LOCK:{tournamentId}
PRIZE_PAYOUT	Tournament Escrow (-) / Winner Balances (+)	TOURNAMENT_SETTLEMENT:{tournamentId}
SHOP_PURCHASE	User Balance (-) / House Treasury (+)	SHOP_ITEM:{itemId}
ADJUSTMENT	Admin Balance Override / Corrective Delta	ADMIN_AUDIT:{adminUserId}

3. Tournament Prize Pool Escrow & Distribution Math

- **Escrow Lock:** When a player joins a paid tournament, entry fee is deducted and locked in tournament escrow.
- **Payout Calculation:** Upon tournament completion:
 - **1 Winner:** 100% of Net Prize Pool.
 - **3 Winners:** 1st: 60%, 2nd: 25%, 3rd: 15%.
 - **Platform Rake:** Configurable 5%–10% deducted into platform treasury.
- **Real-Time Push:** `BalanceGateway` pushes updated balance to connected client sockets immediately upon settlement.